

mind but have beliefs about a whole family of possible probability distributions? For example, an emerging country has newly liberalized. There's no return history. If the country is stable, returns could be drawn from a "good" distribution. But if there are multiple coups, or if investors' capital is confiscated, returns would be drawn from a "bad" distribution. There could be all possible distributions in between. There is a distinction between a single probability distribution—the traditional measure of risk—and a range of possible distributions. We call the latter *Knightian uncertainty*, for Frank Knight (1921), after terminology introduced by Gilboa and Schmeidler (1989).

Under uncertainty aversion, investors' utility depends on both risk and uncertainty. The more precise information is available about the set of possible distributions that can occur, the higher the agent's utility. Agents have an *uncertainty aversion* parameter, just as they have a risk aversion parameter. Uncertainty aversion is also called *ambiguity aversion*, as the existence of multiple distributions is also known as ambiguity. Researchers also talk about *robust utility*, which is closely related and also deals with multiple sets of probability distributions.²⁷ With robust utility, agents exhibit a preference ("concern") for how robust their decisions are under different probability distributions. (My MBA students might call this a glorified version of sensitivity analysis.) It is also referred to as *maxmin utility* because uncertainty averse agents take the worst possible probability distribution of risk (min) and then maximize utility (max).

A comprehensive review of uncertainty aversion and portfolio choice was conducted by Guidolin and Rinaldi (2010). I have the following brief remarks:

1. Uncertainty aversion causes agents to effectively behave in a much more risk-averse fashion than just using plain old risk aversion. You will hold a much smaller weight in equities when you are uncertainty averse. In fact, if you dislike uncertainty enough, you won't hold any equities.²⁸
2. Many of the tools in expected utility can be used—and, more important, the same intuition as in this chapter applies in many cases. There is an explicit connection even to mean-variance utility.²⁹ (Clearly, in a very simple case, expected utility is uncertainty aversion with only one probability distribution.)
3. There are data on agents' different probability beliefs—surveys! Some of the more recent papers in the literature nicely exploit these to show how the set of beliefs evolve over time and how they can be used to infer future asset prices.³⁰

²⁷ The reference tome on robustness is Hansen and Sargent (2007).

²⁸ This was first shown by Dow and Werlang (1992). See also Garlappi, Uppal, and Wang (2007) and Bossaerts et al. (2010), among others.

²⁹ In Maenhout (2004), you can reinterpret the classical CRRA risk aversion coefficient as the sum of the CRRA risk aversion coefficient and the uncertainty risk aversion coefficient. See also Trojani and Vanini (2004).

³⁰ See Ulrich (2011).